



University of Bahrain
College of Information Technology
Department of Computer Engineering

ITCE456-Multimedia Communications

Course Project Proposal

Submitted by:

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Project Title: Audio-Based Text Encoding and Decoding Using Morse Code

Description: This project aims to design and implement a system that converts text into audio using Morse code and then reconstructs the original text from the generated audio signal. The system will first encode input text into Morse code (dots and dashes), and then generate corresponding audio signals where each symbol is represented by tones of different durations separated by silence.

In the decoding stage, the system will analyze the audio signal to detect tone and silence segments, measure their durations, and classify them back into Morse code symbols. These symbols will then be translated back into text.

Additionally, the project will include basic visualizations, such as waveform plots, to illustrate how the audio signal represents the encoded information.

This project demonstrates how information can be represented and converted between different multimedia forms, specifically text and audio.

Objectives:

- Develop a multi-sensor monitoring system for greenhouse conditions
- Enable remote access via a publicly hosted web platform
- Implement secure user authentication and data management
- Add actuator control (e.g., fan/water pump simulation using DC motors)
- Improve system reliability and scalability

Importance: This project demonstrates key concepts in multimedia communications, particularly how information can be represented, encoded, transmitted, and reconstructed across different media formats. It focuses on converting text into audio signals using Morse code and recovering the original message, which illustrates basic principles of signal representation and processing.

The project also highlights fundamental time-domain signal processing concepts such as distinguishing between tone and silence, measuring signal duration, and classifying signals based on timing. These ideas are closely related to real-world communication systems where information is transmitted using timed signals.

Morse code has historically been used in communication systems such as radio and emergency signaling due to its simplicity and reliability, especially in low-bandwidth or noisy environments. It remains relevant in understanding basic digital communication principles and signal robustness.

Finally, the system can serve as a simple educational tool by helping users visually and audibly understand how text is translated into audio signals and back again, reinforcing the relationship between different multimedia representations of the same information.

Work Plan:

1. Literature review and system design
2. Implementation of text-to-Morse encoding and audio generation
3. Implementation of audio-to-Morse decoding and text reconstruction
4. Waveform visualization and basic analysis
5. Testing and refinement of system accuracy
6. Final documentation and presentation preparation

References:

1. J. R. Pierce, An Introduction to Information Theory: Symbols, Signals and Noise, Dover Publications, 1980.
2. S. Haykin, Communication Systems, 5th Edition, Wiley, 2009.
3. National Institute of Standards and Technology (NIST), "Morse Code Signal Representation," <https://www.nist.gov> (accessed 2026).